



WATER MIST SYSTEMS INSPECTION, TESTING, AND MAINTENANCE

Date:	Inspector:	System:	
Location:			
General			
System designation			
Building			
Location of control valve			
Has system been modified since last inspection?	Yes	No	
What is hazard protected?			
Valves			
How are valves supervised?	Sealed	Locked	Tamper switch
Are valves identified with signs?	Yes	No	
Detection System (if any)			
When was the detection system last inspected?			
Operating Instructions			
Are operating instructions present?	Yes	No	
Notes			

WATER MIST SYSTEMS INSPECTION, TESTING, AND MAINTENANCE (Continued)

Name of Property:

Inspector:

Address:

Contract No.:

Property Phone Number:

Date:

Inspection Frequency:	Daily Annual	Weekly Five-Year	Monthly	Quarterly	Semiannual
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Inspections: Daily

Valve (Cold Weather/Heating Season Only)

Yes	No	N/A	Enclosure, not equipped with a low temperature alarm, is inspected during cold weather to verify a minimum temperature of 40°F (4°C)
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Inspections: Weekly

Backflow

Yes	No	N/A	Isolation valves — open position and locked or supervised
Yes	No	N/A	RPA and RPDA — differential-sensing relief valve operating correctly

Control Valves

Yes	No	N/A	In the correct (open or closed) position
Yes	No	N/A	Sealed
Yes	No	N/A	Accessible
Yes	No	N/A	Post Indicator Valves (PIVs) are provided with correct wrenches
Yes	No	N/A	Free from damage or leaks
Yes	No	N/A	Proper signage

Dry/Deluge Valve/Preaction

Yes	No	N/A	Enclosure, where equipped with low temperature alarm, is inspected during cold weather to verify a minimum temperature of 40°F (4°C)
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Master Pressure-Regulating Device

Yes	No	N/A	Downstream pressures are in accordance with design criteria	psi
Yes	No	N/A	Supply pressure is in accordance with design criteria	psi
Yes	No	N/A	Free of damage or leaks	
Yes	No	N/A	Trim in good operating condition	

Inspections: Monthly

Yes	No	N/A	Gauges are in good operating condition	
Yes	No	N/A	Gauges — normal air or nitrogen pressure maintained (unsupervised)	psi
Yes	No	N/A	Gauge on system side of dry valve reads proper ratio of air or nitrogen (unsupervised)	psi
Yes	No	N/A	Gauge on quick-opening device reads the same as system side dry valve gauge (unsupervised)	psi
Yes	No	N/A	Water circulation tank — water level is full (unsupervised)	

Compressed Gas Cylinders

Yes	No	N/A	Cylinder pressure (unsupervised)	psi
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WATER MIST SYSTEMS INSPECTION, TESTING, AND MAINTENANCE (Continued)

Inspections: Monthly

Yes	No	N/A	Compressed gas supply inlet air pressure	psi
Yes	No	N/A	Outlet water pressure	psi
Yes	No	N/A	Inspect and empty the moisture trap, oil injection (pneumatic)	
Yes	No	N/A	Oil level on air regulator unit	

System Control Valves (Locked or Supervised)

Yes	No	N/A	In the correct (open or closed) position	
Yes	No	N/A	Locked or supervised	
Yes	No	N/A	Accessible	
Yes	No	N/A	Post Indicator Valves (PIVs) are provided with correct wrenches	
Yes	No	N/A	Free from damage or leaks	
Yes	No	N/A	Proper signage	

Inspections: Quarterly

Yes	No	N/A	Gauges — normal air or nitrogen pressure maintained when supervised at a constantly attended location	psi
Yes	No	N/A	Gauge on system side of dry valve reads proper ratio of air or nitrogen when supervised at a constantly attended location	psi
Yes	No	N/A	Gauge on quick-opening device reads the same as system side dry valve gauge when supervised at a constantly attended location	psi
Yes	No	N/A	Gauge on supply side of valve reads normal	psi
Yes	No	N/A	Water supply pressure	psi
Yes	No	N/A	Water storage cylinder (high pressure) water level is full (unsupervised)	
Yes	No	N/A	Additive storage cylinder in good condition	
Yes	No	N/A	Water recirculation tank — water level is full (supervised)	
Yes	No	N/A	Compressed gas cylinder properly secured	
Yes	No	N/A	Waterflow alarm and supervisory device is free of damage	
Yes	No	N/A	Hydraulic design information sign is securely attached to riser and legible	

Alarm Valves/Riser Check

Yes	No	N/A	Gauges — normal water pressure maintained	
Yes	No	N/A	Free of damage	
Yes	No	N/A	In appropriate open or closed position	
Yes	No	N/A	Accessible	
Yes	No	N/A	Retard chamber/alarm drains not leaking	

Control Valves (Electronically Supervised)

Yes	No	N/A	In the correct (open or closed) position	
Yes	No	N/A	Electronically Supervised	
Yes	No	N/A	Accessible	

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WATER MIST SYSTEMS INSPECTION, TESTING, AND MAINTENANCE (Continued)

Inspections: Quarterly

Yes	No	N/A
Yes	No	N/A
Yes	No	N/A

Control Valves (Electronically Supervised)

Post Indicator Valves (PIVs) are provided with correct wrenches

Free from damage or leaks

Proper signage

Pressure-Reducing Valve

Yes	No	N/A
Yes	No	N/A
Yes	No	N/A

In the open position and not leaking

Maintaining downstream pressure

In good condition, with hand wheels installed and unbroken

Inspections: Semiannual

Yes	No	N/A
Yes	No	N/A
Yes	No	N/A
Yes	No	N/A
Yes	No	N/A
Yes	No	N/A
Yes	No	N/A

Manually operated indicating valves are in proper position

Pressure in electronically supervised and monitored gas cylinder psi (bar)

Water storage cylinder (high pressure) water level is full (supervised)

Pneumatically operated valves and tubing inspected

Water quality checked (first year)

Additive agent level is full

Enclosure integrity verified

Inspections: Annual

Yes	No	N/A
Yes	No	N/A
Yes	No	N/A

Hoses are not damaged

Water quality checked (after first year)

Low temperature alarm is in good condition

Nozzles

Yes	No	N/A
Yes	No	N/A
Yes	No	N/A
Yes	No	N/A
Yes	No	N/A
Yes	No	N/A

No signs of physical damage, corrosion, or leaking

No loss of fluid in glass bulb heat responsive element

Loading — free of dust

No paint or coating unless applied by the manufacturer

Correct orientation

Correct number and type of spare water mist nozzles

Hangers, Braces, and Support

Yes	No	N/A
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Not damaged, loose, or unattached

WATER MIST SYSTEMS INSPECTION, TESTING, AND MAINTENANCE (Continued)

Inspections: Annual

Yes	No	N/A	High-Pressure Water Storage Cylinder
			Properly secured
Yes	No	N/A	Vent plug is not ruptured or damaged
Yes	No	N/A	Filters are clean
Yes	No	N/A	Inspect cylinder capacity and pressure rating
Yes	No	N/A	Check cylinder compliance specification
Yes	No	N/A	Cylinder pressure on discharge psi (bar)
Yes	No	N/A	Water recirculation tank supports and attachments in good condition
Yes	No	N/A	Water recirculation tank clean filters, strainers, and cyclone separator

Piping, Tubing, and Fittings

Yes	No	N/A	Free of mechanical damage
Yes	No	N/A	Free of leakage
Yes	No	N/A	Free of external loads
Yes	No	N/A	Missing or damaged paint or coatings
Yes	No	N/A	Free of corrosion or rust
Yes	No	N/A	Misalignment or trapped sections
Yes	No	N/A	Low point drains not damaged or corroded

Compressed Gas Cylinders

Yes	No	N/A	Inspect cylinder capacity and pressure rating
Yes	No	N/A	Check cylinder compliance specification
Yes	No	N/A	Compressed gas meets specifications

Inspection: Five Years

Yes	No	N/A	Alarm valve — interior including strainers, filters, and restricted orifices
Yes	No	N/A	Obstruction inspection — no foreign or obstructing material found
Yes	No	N/A	Check valve — internal moves freely and in good condition
Yes	No	N/A	Backflow — internal inspection
Yes	No	N/A	Compressed gas cylinder exterior inspection

Test: Quarterly

Yes	No	N/A	Water mist system operated by pressurized gas cylinders or pressurized vessels, nozzles flow upon operation
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Test: Semiannual

Yes	No	N/A	Alarm device — vane and pressure switch type
Yes	No	N/A	Valve supervisory switch(es) function
Yes	No	N/A	Pneumatic slave valve and cycle on off valve test

WATER MIST SYSTEMS INSPECTION, TESTING, AND MAINTENANCE (Continued)

Test: Annual

Yes	No	N/A	Supervisory switch(s) function
Yes	No	N/A	Water supply tested
			psi gpm minute duration
Yes	No	N/A	Additive agent quality tested
Yes	No	N/A	Additive agent injection — full discharge test
Yes	No	N/A	Valve status test
Yes	No	N/A	Standby pump operated
Yes	No	N/A	All control valves operated through full range of motion and returned to normal position

Backflow

Yes	No	N/A	Forward flow test at a minimum flow rate of the system demand
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Master Pressure-Regulating Valve

Yes	No	N/A	Full flow test compared to previous test results
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Water Recirculation Tank

Yes	No	N/A	Operational test per manufacturers instructions
Yes	No	N/A	Float-operated valve functions
Yes	No	N/A	Pressure at outlet during discharge psi
Yes	No	N/A	Backflow preventer functions

Pneumatic Valves

Yes	No	N/A	Solenoid release for electronic and manual release
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Full Flow Trip Test

Yes	No	N/A	Unobstructed discharge from all nozzles
Yes	No	N/A	Pressure reading at control valve psi
Yes	No	N/A	Detection system response time sec
Yes	No	N/A	Detection system tested in accordance with NFPA 72®
Yes	No	N/A	Ventilation system interlocks tested
Yes	No	N/A	Fuel/lubrication system interlocks tested

Test: Five Years

Yes	No	N/A	Gauges tested or replaced
Yes	No	N/A	DOT, CTC, or similar compressed gas/water cylinder retesting
Yes	No	N/A	Hose hydrostatic test

WATER MIST SYSTEMS INSPECTION, TESTING, AND MAINTENANCE (Continued)

Routine Maintenance

Yes	No	N/A	OS&Y — stems lubricated annually
Yes	No	N/A	Flush mainline strainer after each test
Yes	No	N/A	Evaluate additives per manufacturer
Yes	No	N/A	Drain and refill water tank annually
Yes	No	N/A	Rebuild pneumatically operated standby pump every 5 years
Yes	No	N/A	Inspect representative sample of operated water mist nozzles after activation

Comments

Signature:

Date:

Contractor Name:

Contractor Address:

License/Certification No.:

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WATER MIST SYSTEMS INSPECTION, TESTING, AND MAINTENANCE (Continued)

Name of Property:

Inspector:

Address:

Contract No.:

Property Phone Number:

Date:

Inspection Frequency: **Weekly** **Monthly** **Quarterly** **Annual** **Other**

This form covers weekly inspections over a 6-month period.

Year:

System:

Location:

General:

1. Check that water tank is full if water level not supervised.
2. Check air pressure gauge to confirm that adequate air pressure is being maintained.
3. Check air pressure gauge to confirm that adequate air pressure is being maintained by the compressor.
4. Record any notes about the system that the inspector believes to be significant.

Y = Satisfactory

N = Unsatisfactory (explain below)

Date	Inspector	Water Level (1)	Air Tank Pressure (2)	Air Compressor (3)	Notes (4)

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